

Stroke Prediction: Data Preprocessing Proposal

ITAI 1371 Midterm Project | Team ML_1371_12321 (Edwin, Kala, Anashrah, Zain) | Instructor: Mr. Rao

1. The Dataset

We use the **Stroke Prediction Dataset** (Kaggle, published by fedesoriano): **5,110 rows and 12 columns**, each row a unique patient. Features include demographic and health attributes such as age, gender, hypertension, heart disease, marital status, work type, residence type, average glucose level, BMI, and smoking status. The target column, **stroke**, is binary (1 = the patient had a stroke, 0 = no stroke). Source: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>

2. Problem Statement

This is a **supervised binary classification** problem: predict whether a patient is likely to experience a stroke from their demographic and health profile. It fits our team's healthcare focus and gives us a clinically meaningful task to build models around in the finals.

3. Why This Dataset

Beyond domain relevance, the dataset presents the exact data challenges this course asks us to handle: **missing values** in the BMI column, a mix of **categorical and numeric** features that require encoding and scaling, and a **severe class imbalance**, since only 249 of 5,110 patients (about 5%) had a stroke. A model trained on the raw imbalance could predict 'no stroke' for everyone and still appear about 95% accurate while catching no actual strokes, which is why balancing is essential.

4. Preprocessing Plan

All work is done in Python within a Jupyter or Colab notebook, and the raw file is never edited manually. We follow a pipeline built to avoid data leakage, in this order:

- **Stratified 70/30 split first**, so both the training and test sets keep the real stroke rate of about 5%. The test set is then set aside and never touched.
- **EDA on the training data only**, including summary statistics, a missing value check, and distribution and class balance plots.
- **Median imputation** for the missing BMI values, then encoding of the categorical columns and scaling of the numeric features, all fit on the training set and then applied to the test set.
- **SMOTE applied to the training set only** to balance the classes, with before and after Matplotlib plots. The test set keeps its real stroke rate for honest evaluation.

5. What We Will Accomplish

The midterm deliverable is a clean, balanced training set that is ready for modeling, plus a documented notebook showing the data before and after each preprocessing step. For the finals, we will train and compare several classifiers, starting with a Logistic Regression baseline and then models based on decision trees, using this prepared data. Because the problem is imbalanced and medical, we will evaluate mainly on recall or sensitivity for the stroke class and on ROC AUC rather than raw accuracy, since missing a real stroke is far costlier than a false alarm. We will also examine the data for possible fairness and bias concerns across demographic groups.